MILITARY STANDARD

MIL-STD-1320-164
(NAVY)

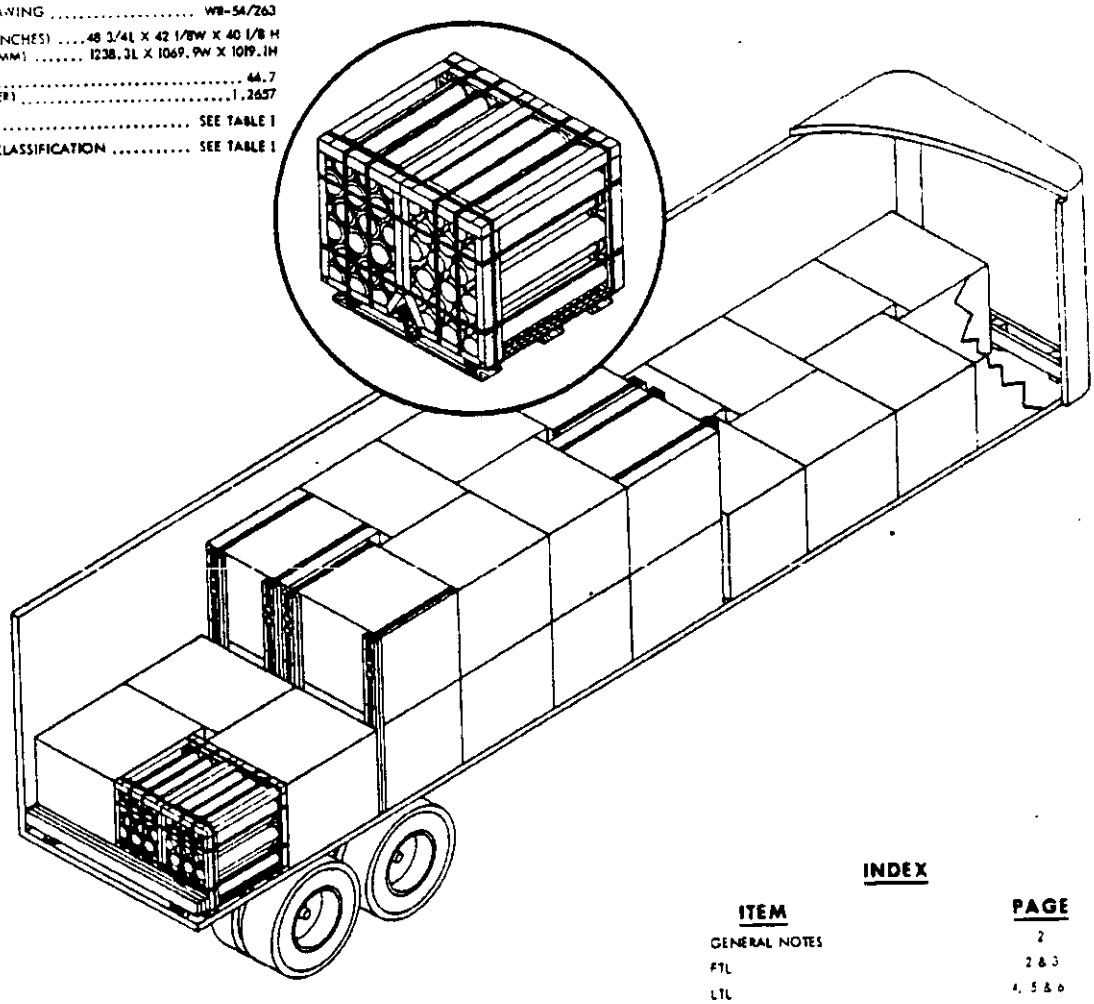
28 FEBRUARY 1975

TRUCKLOADING OF HAZARDOUS MATERIALS

CARTRIDGE, 76 MM/62 CALIBER
CHARGE, CLEANING, 76 MM/62 CALIBER
IN PALLET ADAPTER MK 121 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WB-54/263
DIMENSIONS (INCHES) ....48 3/4L X 42 1/8W X 40 1/8 H
(MM) ..... 1238.3L X 1069.9W X 1019.1H
CUBE (CU FT) 44.7
(CU METER) 1.2657
WEIGHT ..... SEE TABLE I
DOT HAZARD CLASSIFICATION ..... SEE TABLE I



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236.

INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL	2 & 3
LTL	4, 5 & 6
UNITIZING	7

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M J Gray NWHL 2/25/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J F Lee 2/25/75
SIGNATURE NAVSEASYSCOM BY DIRECTION DATE

ORIGINATOR
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING 76 MM 62 CALIBER CARTRIDGES AND OR CLEARING CHARGES WHEN THEY ARE PALLETIZED IN UNIT LOADS IN ACCORDANCE WITH WR-54-263.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42 AND 44 FT VAN TYPE TRAILERS WITH THE TRAILER TANDEM AXLES LOCATED IN THE 'WESTERN' POSITION (AT THE EXTREME REAR OF THE TRAILER). THE TRAILERS MAY HAVE WOOD FLOORS, METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS.

WARNING

DO NOT NAIL BLOCKING OR BRACING INTO METAL FLOORS.

3. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND LAYER WHERE THE STACK CHANGES FROM TWO LAYERS TO ONE LAYER HIGH, THE UNIT LOADS SHALL BE UNITIZED AS SHOWN ON PAGE 7. THE STIFFENERS SHALL BE POSITIONED ON THE UNIT LOADS SO THAT WHEN THE UNIT LOADS ARE IN THE TRAILER, THE STRAPPING IS PARALLEL TO THE LONGITUDINAL AXIS OF THE TRAILER. THE STIFFENERS SHALL ALSO BE POSITIONED TOWARDS THE UNIT LOADS THAT ARE ONE HIGH. IF BOTH ENDS OF THE STACKED LOADS ARE AGAINST ONE HIGH UNIT LOADS, STIFFENERS SHALL BE USED ON BOTH ENDS OF THE STACKED TWO HIGH LOADS.
4. ONE OR MORE UNIT LOADS OF CLEARING CHARGES MAY BE SUBSTITUTED IN A FTL OR LTL IF REQUIRED.
5. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
6. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH APPROPRIATE EXPLOSIVES PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER. (SEE TABLE I.)
7. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
8. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40,42 AND 44 FT TRAILERS (VANS)

1. A FTL CONSISTS OF 28 UNIT LOADS, LOADED, BLOCKED AND BRACED AS SHOWN ON PAGE 3.
2. THE METHOD OF REAR BLOCKING TO BE USED IS DEPENDENT UPON THE LENGTH OF THE TRAILER.

   FOR 40 FT TRAILERS

   INSTALL REAR BLOCKING ASSEMBLY, DETAIL A, AS SHOWN ON PAGE 3.

   WARNING

   REAR BLOCKING MUST BEAR AGAINST THE LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

   FOR 42 FT TRAILERS

   INSTALL REAR BLOCKING ASSEMBLY, DETAIL D, AS SHOWN ON PAGE 5.

   CAUTION

   REAR BLOCKING MUST BEAR AGAINST THE LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

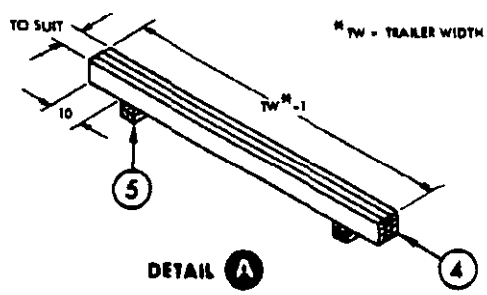
   FOR 44 FT TRAILERS

   INSTALL REAR BLOCKING AS SHOWN IN TYPICAL LTL, PAGE 6.

   WARNING

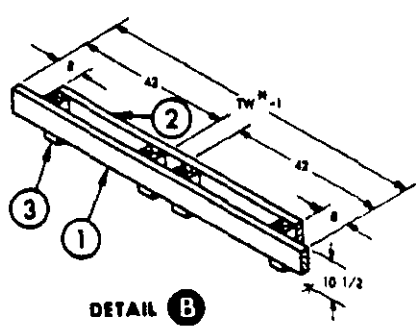
   DO NOT NAIL BLOCKING INTO METAL FLOORS. IF FLOORS ARE ALL METAL, USE METHOD FOR 42 FT TRAILERS.

3. WHEN LOADING A 44 FT TRAILER, USE FRONT BULKHEAD, DETAIL H PAGE 6, TO OBTAIN PROPER WEIGHT DISTRIBUTION.



DETAIL A

REAR BLOCKING ASSEMBLY



DETAIL B

FRONT BULK
1 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:45Z
Check the source to verify that this is the current version before use.

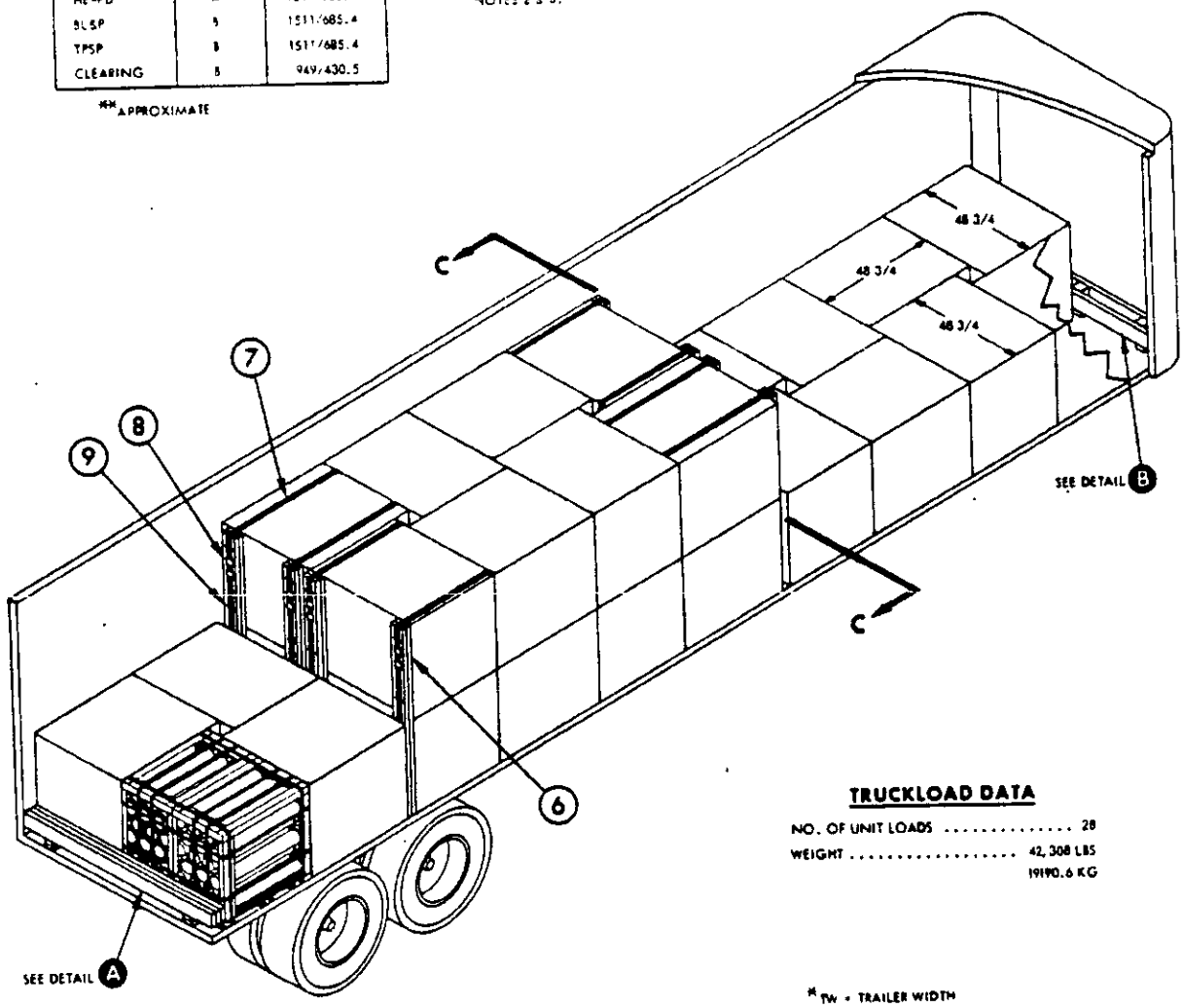
TABLE I

	HAZARD CLASS	UNIT LOAD WT. LBS/KG
HE-IR	A	1511/685.4
HE-PD	A	1511/685.4
BLSP	B	1511/685.4
TPSP	B	1511/685.4
CLEARING	B	949/430.5

**APPROXIMATE

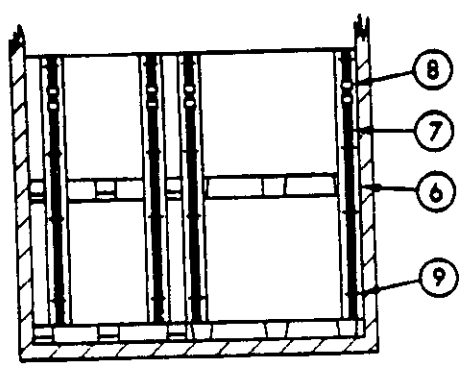
NOTE

TRUCKLOAD SHOWN IS FOR 40 FT TRAILERS. WHEN LOADING 42 & 44 FT TRAILERS, ADJUST BLOCKING AS INSTRUCED BY FTL NOTES 2 & 3.



TRUCKLOAD DATA

NO. OF UNIT LOADS 28
WEIGHT ..... 42,308 LBS
19190.6 KG



SECTION C-C

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
9	STAPLE	FOR 1 1/4 STRAP	32	-	-	-
8	SEAL	FOR 1 1/4 STRAP	16	-	-	-
7	STRAP	1 1/4 x .035 x 23 FT	8	-	-	-
6	STIFFENER	2 x 6 x 75 1/2	16	DOUBLE LAMINATED	5	10d
				4	2/JOINT	10d
5	SUPPORT	2 x 4 TO SUIT	4	5	3	10d
4	CROSSMEMBER	THICKNESS TO SUIT x 6 x TW*-1	AS REQD	LAMINATE	6/PIECE	TO SUIT
3	VERTICAL	4 x 4 x 10 1/2	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 x 6 TO SUIT	1	3	3/JOINT	16d
1	AFT HORIZONTAL	2 x 6 x TW*-1	1	3	3/JOINT	16d

LIST OF MATERIALS AND NAILING DATA

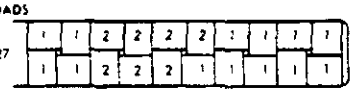
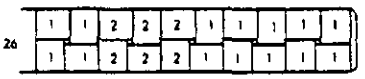
FTL TRAILER (VAN)

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:45Z
Check the source to verify that this is the current version before use.

LTL 40,42 AND 44 FT TRAILERS (VANS)

1. WHEN REQUIRED TO SHIP 16 TO 27 UNIT LOADS, SELECT THE CORRECT LOAD PATTERN FROM TABLE II. LOAD THE TRAILER USING THIS PATTERN AND BLOCK AND BRACE THE LOAD USING THE PRINCIPLES OF THE LTL TRUCKLOADS ON PAGES 5 AND 6.

TABLE II

NO. OF UNIT LOADS	LOAD PATTERN*	FWD
27		
26		
25		
24		
23		
22		
21		
20		
19		
18		
17		
16		

*LOAD PATTERN SHOWS TRAILER FLOOR PLAN. NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN LAYER.

2. WHEN REQUIRED TO SHIP LESS THAN 16 UNIT LOADS FOLLOW THE BASIC LOAD PATTERN FOR 16 TO 19 UNIT LOADS SHOWN IN TABLE II. BLOCK AND BRACE THE LOAD USING THE BASIC PRINCIPLES OF THE LTL LOAD SHOWN ON PAGE 6. THE REAR BLOCKING MAY BE STAGGERED TO ACCOMODATE LOAD CONFIGURATION.

WARNING

DO NOT USE TRAILERS WITH ALL METAL FLOORS FOR LTL SHIPMENTS AS THEY CAN NOT BE BLOCKED BY THIS METHOD. USE TRAILERS WITH WOOD FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS.

3. WHEN A UNIT LOAD IN THE SECOND LAYER DOES NOT HAVE A UNIT LOAD OPPOSITE IT, USE SWAY BRACE ASSEMBLY, DETAIL E, ADJACENT TO IT. SECURE ASSEMBLY IN PLACE BY TIEING TO STRAPS, PIECE 13, WITH SOFT ANNEALED IRON WIRE.
4. WHEN LOADING A 44 FT TRAILER WITH 25, 26 OR 27 UNIT LOADS USE FRONT BULKHEAD, DETAIL H PAGE 6, TO OBTAIN PROPER WEIGHT DISTRIBUTION.
5. THE METHOD OR REAR BLOCKING TO BE USED IS DEPENDENT UPON THE LENGTH OF THE TRAILER.

FOR 40 FT TRAILERS

INSTALL REAR BLOCKING ASSEMBLY, DETAIL A. AS SHOWN ON PAGE 3.

CAUTION

REAR BLOCKING MUST BEAR AGAINST THE LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED.
DO NOT USE TRAILERS WITH ROLL UP DOORS.

FOR 42 FT TRAILERS

INSTALL REAR BLOCKING ASSEMBLY, DETAIL D AS SHOWN ON PAGE 5.

CAUTION

REAR BLOCKING ASSEMBLY MUST BEAR AGAINST THE LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED.
DO NOT USE TRAILERS WITH ROLL UP DOORS.

FOR 44 FT TRAILERS

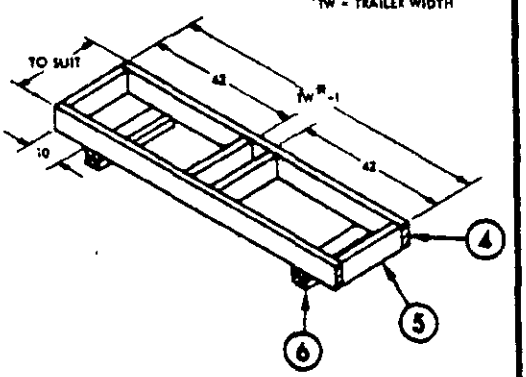
INSTALL REAR BLOCKING AS SHOWN IN TYPICAL LTL, PAGE 6.

WARNING

DO NOT NAIL BLOCKING INTO METAL FLOORS. IF FLOORS ARE ALL METAL, USE METHOD FOR 42 FT TRAILER.

6. THE LOAD SHOWN ON PAGE 5 IS FOR 23 UNIT LOADS AND THE LOAD ON PAGE 6 FOR 6 UNIT LOADS. THEY ARE INTENDED TO ILLUSTRATE THE TYPICAL BLOCKING AND BRACING FOR LTL LOADS AND DO NOT LIMIT LTL LOADS TO THESE QUANTITIES.

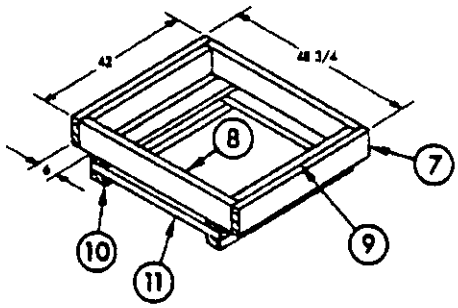
*TW = TRAILER WIDTH



DETAIL (D)

REAR BLOCKING ASSEMBLY
1 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.

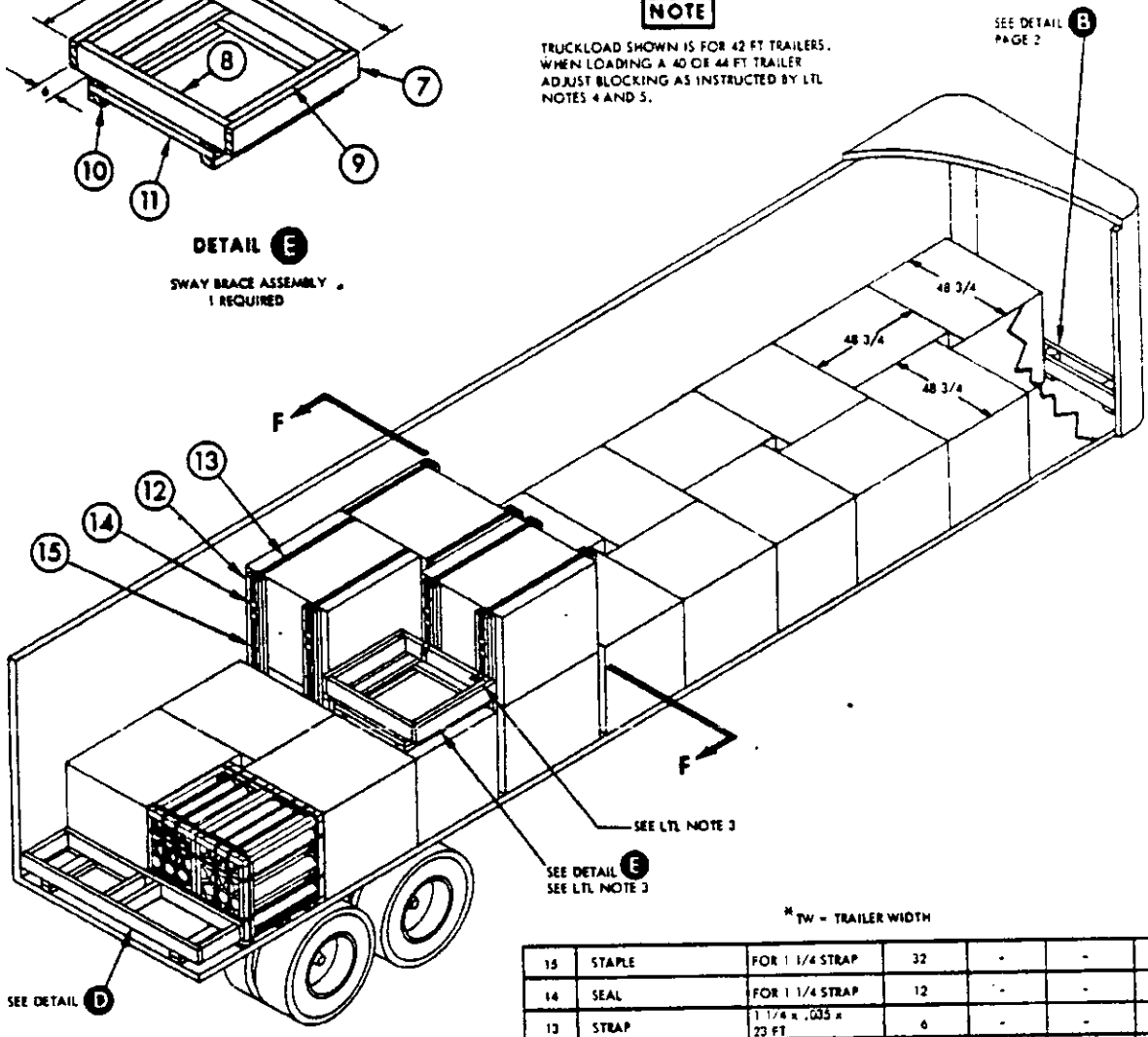


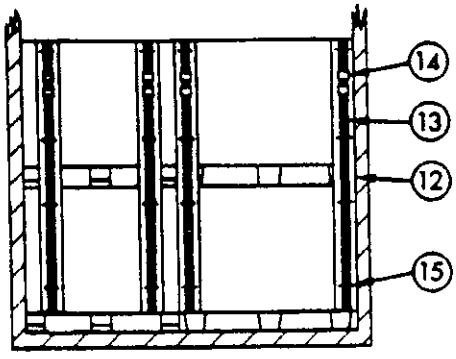
DETAIL E

SWAY BRACE ASSEMBLY
1 REQUIRED

NOTE

TRUCKLOAD SHOWN IS FOR 42 FT TRAILERS. WHEN LOADING A 40 OR 44 FT TRAILER ADJUST BLOCKING AS INSTRUCTED BY LTL NOTES 4 AND 5.





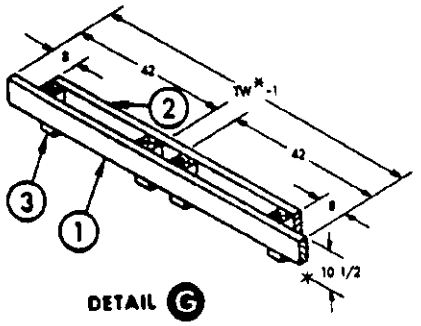
SECTION F-F

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
15	STAPLE	FOR 1 1/4 STRAP	32	-	-	-
14	SEAL	FOR 1 1/4 STRAP	12	-	-	-
13	STRAP	1 1/4 x .035 x 23 FT	6	-	-	-
12	STIFFENER	2 x 6 x 75 1/2	16	DOUBLE LAMINATE	5	10d
11	TIE	2 x 4 x 36	2	10	2 JOINT	12d
10	SUPPORT	2 x 4 x 42	4	8 11	2 JOINT 2 JOINT	12d 12d
9	STRUT	2 x 6 x 39	2	SEE 8	-	-
8	LONGITUDINAL	2 x 6 x 45 3/4	2	9	3 JOINT	16d
7	CROSSMEMBER	2 x 6 x 42	2	8	3 JOINT	16d
6	SUPPORT	2 x 4 TO SUIT	4	4 6	2 JOINT 3	10d 10d
5	STRUT	2 x 6 TO SUIT	4	SEE 4	-	-
4	CROSSMEMBER	2 x 6 x TW* -1	2	5	3 JOINT	16d
3	VERTICAL	4 x 4 x 10 1/2	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 x 6 TO SUIT	1	3	3 JOINT	16d
1	AFT HORIZONTAL	2 x 6 x TW* -1	1	3	3 JOINT	16d

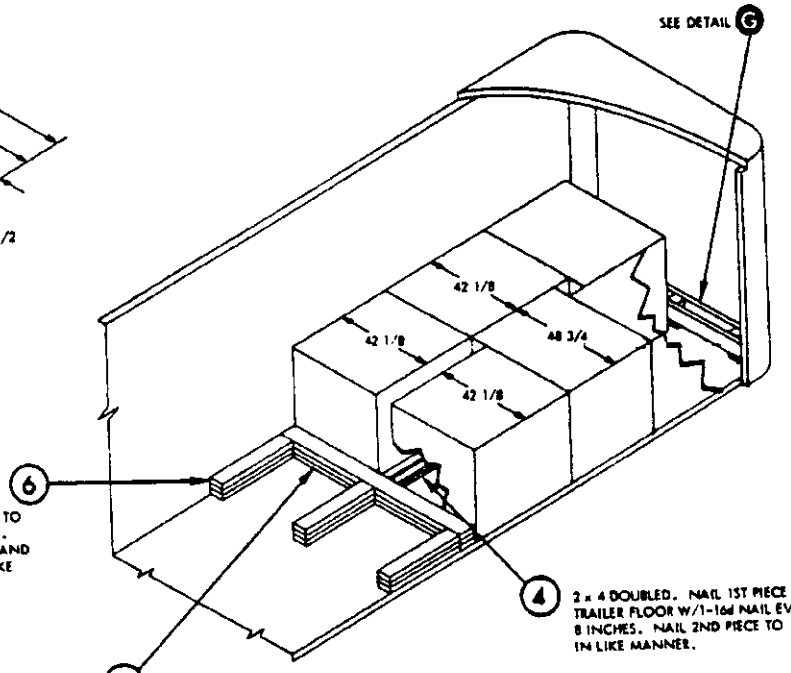
LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.



DETAIL G

FRONT BULK
1 REQUIRED



2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/9-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

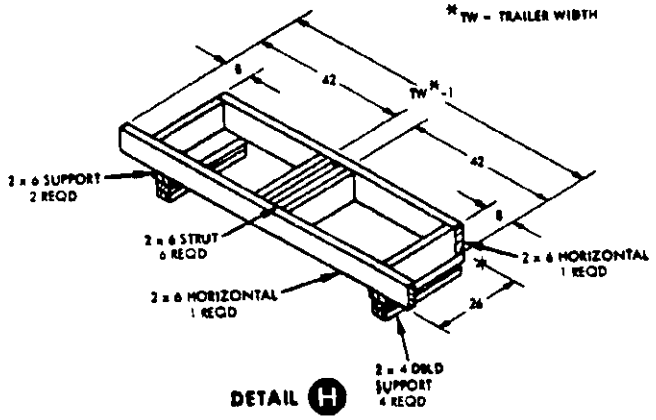
*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS	
					NUMBER	SIZE
6	BACKUP CLEAT	2 x 6 x 36	9	SEE FIELD NOTE		
5	CROSSMEMBER	2 x 6 x TW*-1	3	SEE FIELD NOTE		
4	SLEEPER	2 x 4 x 48	4	SEE FIELD NOTE		
3	VERTICAL	4 x 4 x 10 1/2	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 x 6 TO SUIT	1	3	3/JOINT	16d
1	AFT HORIZONTAL	2 x 6 x TW*-1	1	3	3/JOINT	16d

LIST OF MATERIALS AND NAILING DATA

LTL VAN

*TW = TRAILER WIDTH

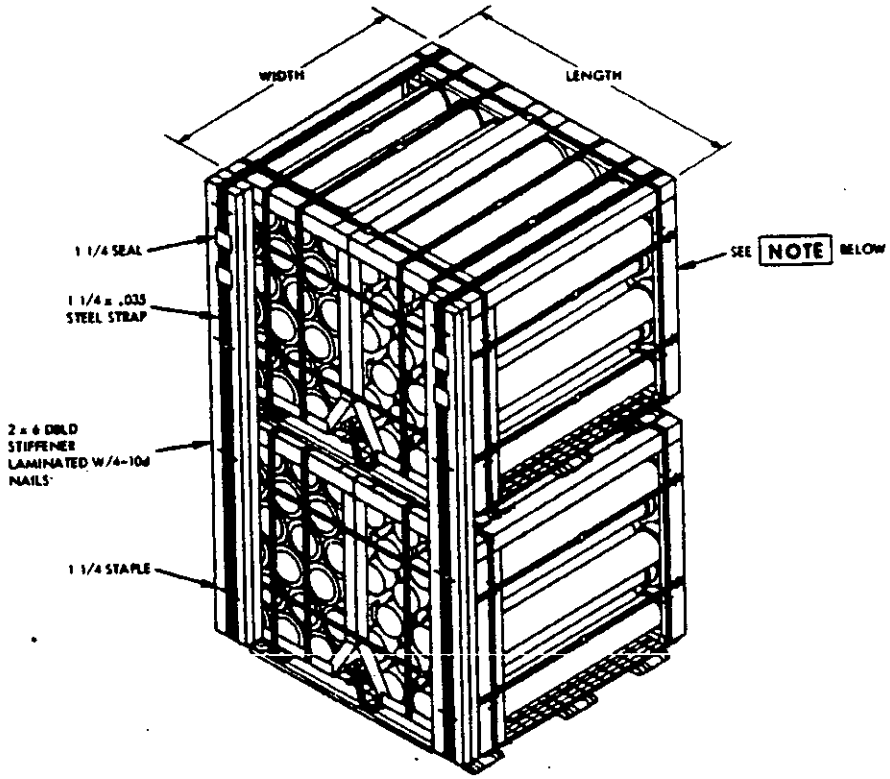


DETAIL H

FRONT BULKHEAD
FOR 40 FT TRAILER
SEE FTL NOTE 3 & LTL NOTE 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.

UNITIZING DETAIL



1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, STACKED UNIT LOADS SHALL BE UNITIZED.

WARNING

THE CHIMNEY PATTERN REQUIRES UNITIZING IN TWO CONFIGURATIONS WITH THE STIFFENERS ON THE "LENGTH" SIDE OF THE STACKED UNIT LOADS AS SHOWN ABOVE AND STIFFENERS ON THE "WIDTH" SIDE. UNITIZING SHALL BE ACCOMPLISHED SO THAT THE REQUIREMENTS OF GENERAL NOTE 3, PAGE 2 ARE SATISFIED.

2. THE DOUBLED 2 x 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD.
3. THE 1 1/4 INCH STEEL STRAPS POSITIONED AS SHOWN, ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE TOP DECK OF THE BOTTOM PALLET.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STRAP STAPLES.

NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACK OF UNIT LOADS, THE STIFFENERS SHALL BE AT BOTH SIDES.

5. WHEN UNITIZING THREE HIGH, FOLLOW THE PRINCIPLES OF THE TWO HIGH UNIT LOADS SHOWN ABOVE.

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N283)

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:45Z
Check the source to verify that this is the current version before use.